

NEW CASTLE, PA, USA

WMO: 720378

Lat: 41.033N

Lon: 80.417W

Elev: 327

StdP: 97.46

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 00122

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.6	-14.5	-22.0	0.5	-16.2	-19.0	0.7	-13.6	10.3	-0.2	8.8	-1.2	2.0	250	0.462

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	32.2	23.5	30.9	22.9	29.1	21.9	24.9	29.8	24.0	28.4	23.2	27.2	3.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.5	19.0	27.1	22.6	18.1	26.0	22.1	17.4	25.5	77.2	29.8	73.4	28.5	70.3	26.9	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.9	7.5	6.4	-22.0	33.7	3.1	1.5	-24.3	34.9	-26.1	35.8	-27.9	36.6	-30.1	37.7		
(5)			-21.9	26.3	3.7	0.9	-24.5	26.9	-26.7	27.4	-28.7	27.9	-31.4	28.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.2	-3.4	-2.3	3.2	9.9	16.0	19.9	22.5	21.4	18.0	11.6	4.9	0.5
	DBStd	10.37	6.81	6.84	6.40	5.55	4.85	3.52	2.91	2.85	3.99	5.08	5.21	5.37
	HDD10.0	1576	417	348	230	70	7	0	0	0	0	40	167	296
	HDD18.3	3346	673	578	471	258	103	24	3	7	54	219	403	552
	CDD10.0	1665	3	3	18	66	194	297	386	352	240	88	14	2
	CDD18.3	393	0	0	0	4	32	71	132	100	44	9	1	0
	CDH23.3	3634	0	0	5	71	370	651	1221	835	409	71	1	0
	CDH26.7	1133	0	0	1	12	99	196	438	243	130	15	0	0
Wind	WSAvg	2.4	3.0	2.8	2.9	3.0	2.2	2.0	1.7	1.5	1.7	2.4	2.6	2.8
Precipitation	PrecAvg	935	61	51	73	80	90	97	103	87	89	67	77	67
	PrecMax	1390	178	106	141	158	195	214	268	178	277	146	259	145
	PrecMin	702	16	5	23	18	23	17	31	32	15	8	18	17
	PrecStd	162	32	22	28	34	40	45	53	41	48	34	38	25
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.4	18.8	23.8	28.0	31.4	32.6	33.9	32.4	32.4	28.9	21.3	17.2
		MCWB	13.4	13.7	15.6	16.9	21.6	23.4	24.2	24.0	23.2	22.2	14.3	14.0
	2%	DB	13.1	14.2	19.3	25.8	29.1	31.0	32.4	31.0	30.3	25.1	18.1	13.9
		MCWB	11.4	10.8	13.0	16.0	20.5	22.6	23.7	23.3	22.9	18.4	12.7	11.5
	5%	DB	9.5	11.3	17.0	22.6	27.5	28.9	31.0	29.1	27.8	22.7	16.2	12.0
		MCWB	7.7	8.9	11.5	14.4	19.3	21.7	23.3	22.2	21.3	17.4	11.9	10.0
	10%	DB	6.7	7.7	13.8	19.7	25.4	27.3	29.0	27.7	25.8	20.1	13.5	8.8
		MCWB	4.6	5.2	9.7	12.4	18.3	20.3	22.1	21.2	20.2	16.6	9.9	7.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.7	14.6	16.6	18.5	23.2	24.7	26.5	25.7	25.1	22.7	16.6	14.8
		MCDB	15.5	17.7	22.2	25.5	28.7	30.0	31.5	30.5	30.5	27.6	19.5	16.4
	2%	WB	11.7	11.8	14.7	17.1	21.8	23.7	25.3	24.4	23.7	20.4	14.2	12.3
		MCDB	13.3	13.3	18.8	23.5	27.2	28.9	30.4	28.6	28.0	23.0	16.6	13.6
	5%	WB	8.0	8.8	12.6	15.6	20.6	22.7	24.2	23.5	22.4	18.6	12.3	9.9
		MCDB	9.1	11.7	15.7	20.7	25.7	27.4	28.7	27.2	26.3	21.0	15.4	11.4
	10%	WB	4.5	5.2	10.1	14.0	19.4	21.6	23.3	22.6	21.2	17.0	10.3	6.9
		MCDB	6.6	7.7	12.9	18.7	23.7	25.5	27.2	26.0	24.4	19.6	13.1	8.7
Mean Daily Temperature Range	5% DB	MDBR	8.5	9.2	10.4	12.3	12.8	12.0	12.1	11.9	11.9	10.8	9.6	7.6
		MCDBR	11.0	13.2	14.1	16.7	15.3	13.9	13.7	13.3	14.3	14.5	13.7	11.3
		MCWBR	9.7	10.6	9.2	9.4	7.5	6.7	6.1	6.3	7.1	8.1	9.0	9.6
	5% WB	MCDBR	10.3	12.7	12.2	14.0	13.1	11.8	11.7	10.9	12.2	11.5	11.6	10.7
		MCWBR	9.7	10.8	8.9	9.0	6.8	6.3	5.8	5.5	6.4	7.4	8.9	10.2
Clear-Sky Solar Irradiance	taub	0.301	0.330	0.355	0.383	0.421	0.449	0.451	0.442	0.386	0.361	0.318	0.301	
	taud	2.439	2.339	2.350	2.310	2.218	2.177	2.190	2.207	2.340	2.406	2.508	2.490	
	Ebn at Noon	856	876	893	888	858	831	825	822	849	830	832	829	
	Edh at Noon	78	101	112	125	140	146	143	137	112	94	73	69	
All-Sky Solar Radiation	RadAvg	1.44	2.31	3.27	4.45	5.25	5.72	5.77	5.18	4.19	2.73	1.81	1.17	
	RadStd	0.14	0.29	0.33	0.46	0.46	0.48	0.43	0.29	0.42	0.26	0.25	0.12	

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (40 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+116	+56

Nomenclature: See separate page